

C263F PROJECT PROPOSAL

Vibration and Damping Analysis of a Bridge Stay Cable

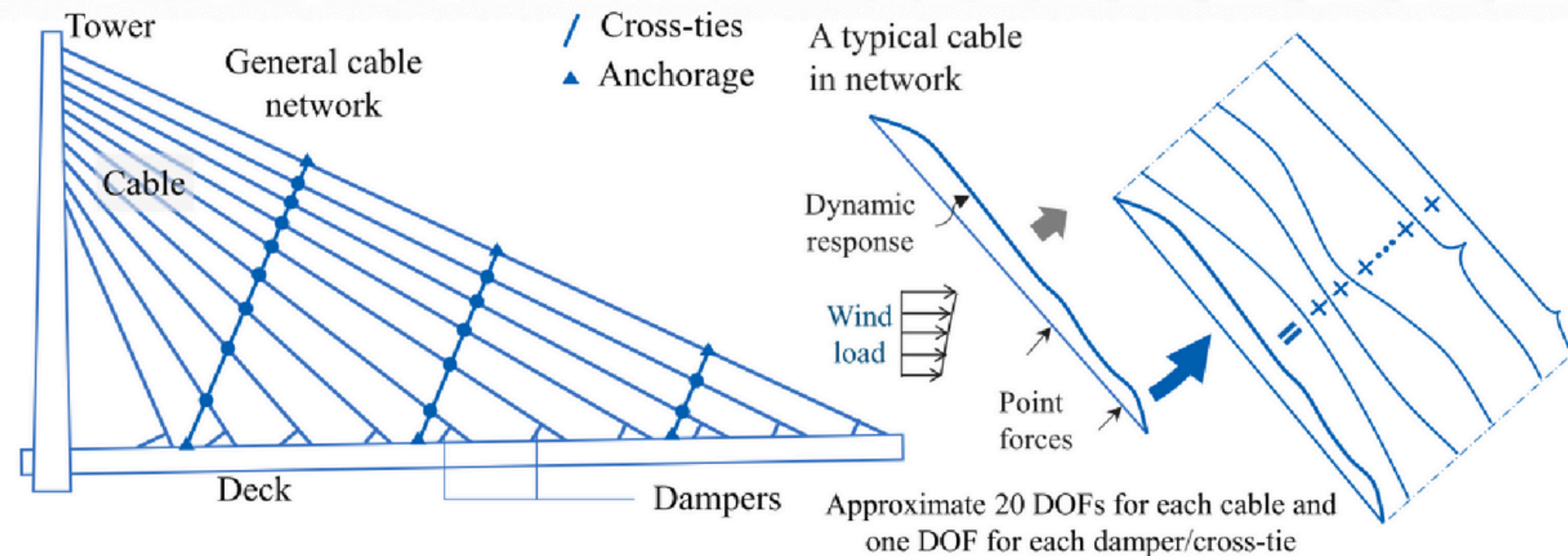
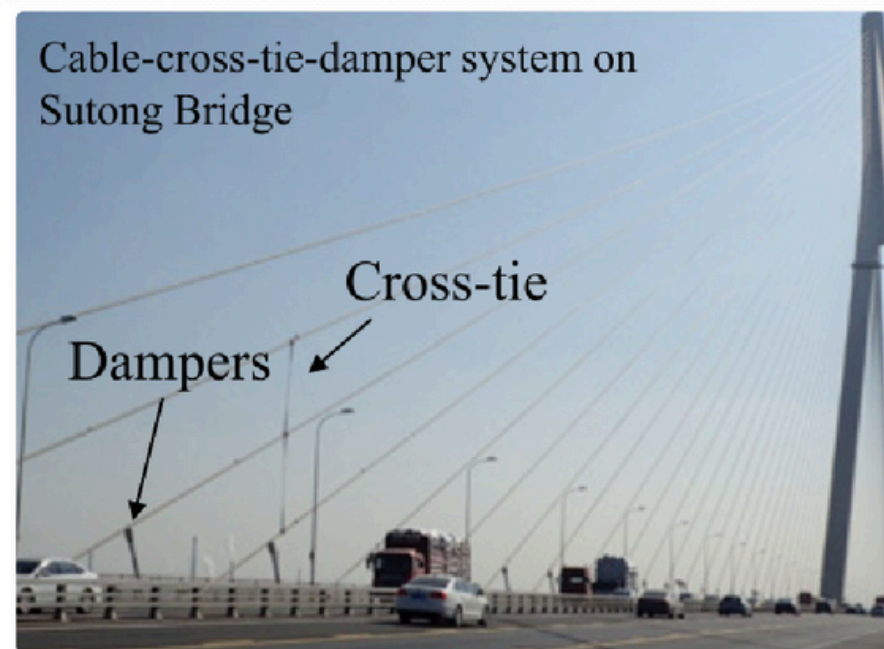
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Project Overview

A brief summary of the project content

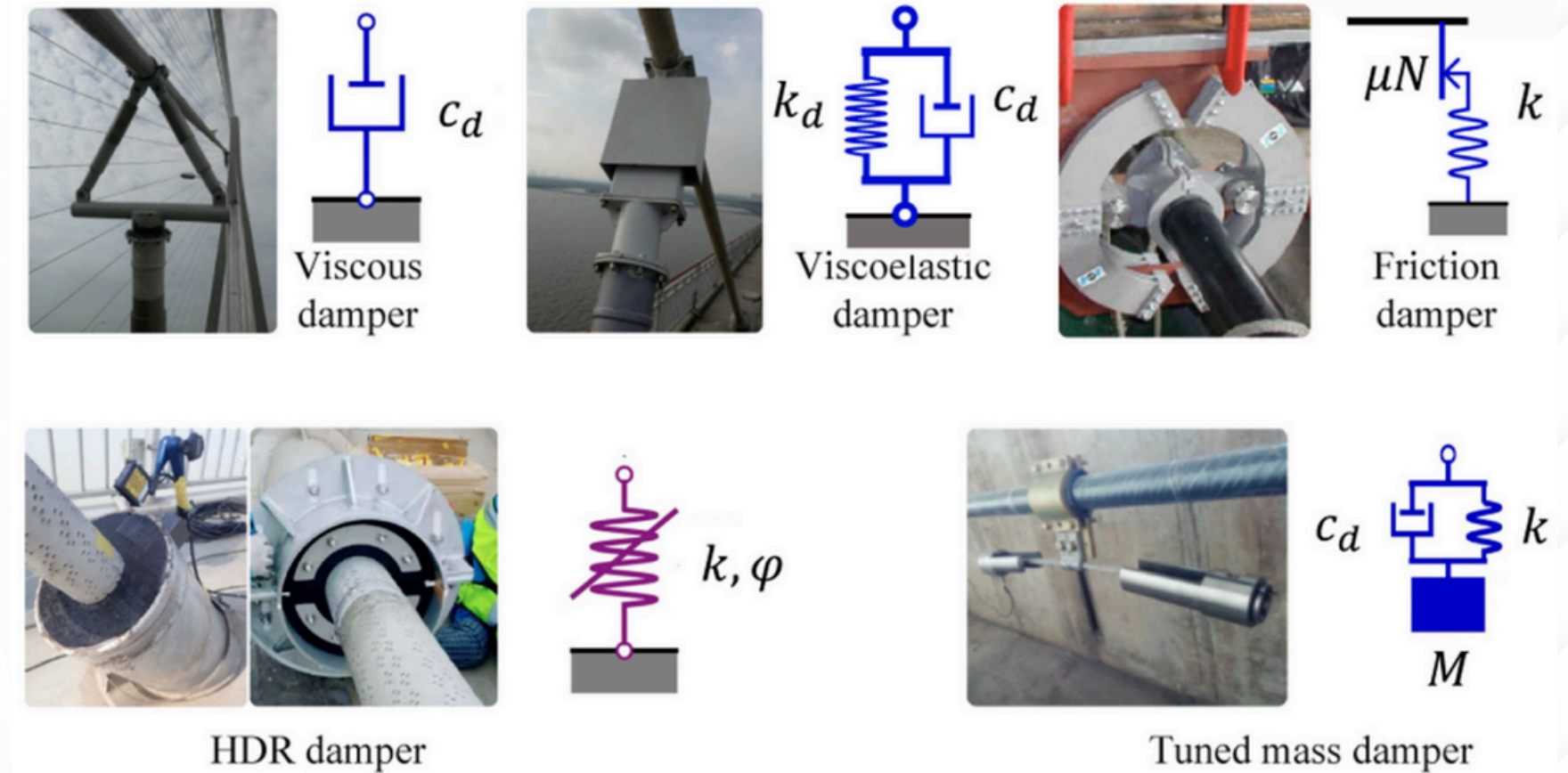
- **Objective:** Develop a discrete-node model to study vibration and damping of a single inclined stay cable.
- **Motivation:** Stay cables in cable-stayed bridges are prone to wind-, rain-, and traffic-induced vibrations that compromise fatigue life.
- **Approach:** Represent the cable as multiple nodes linked by springs and dampers, capturing geometric nonlinearity and energy dissipation.



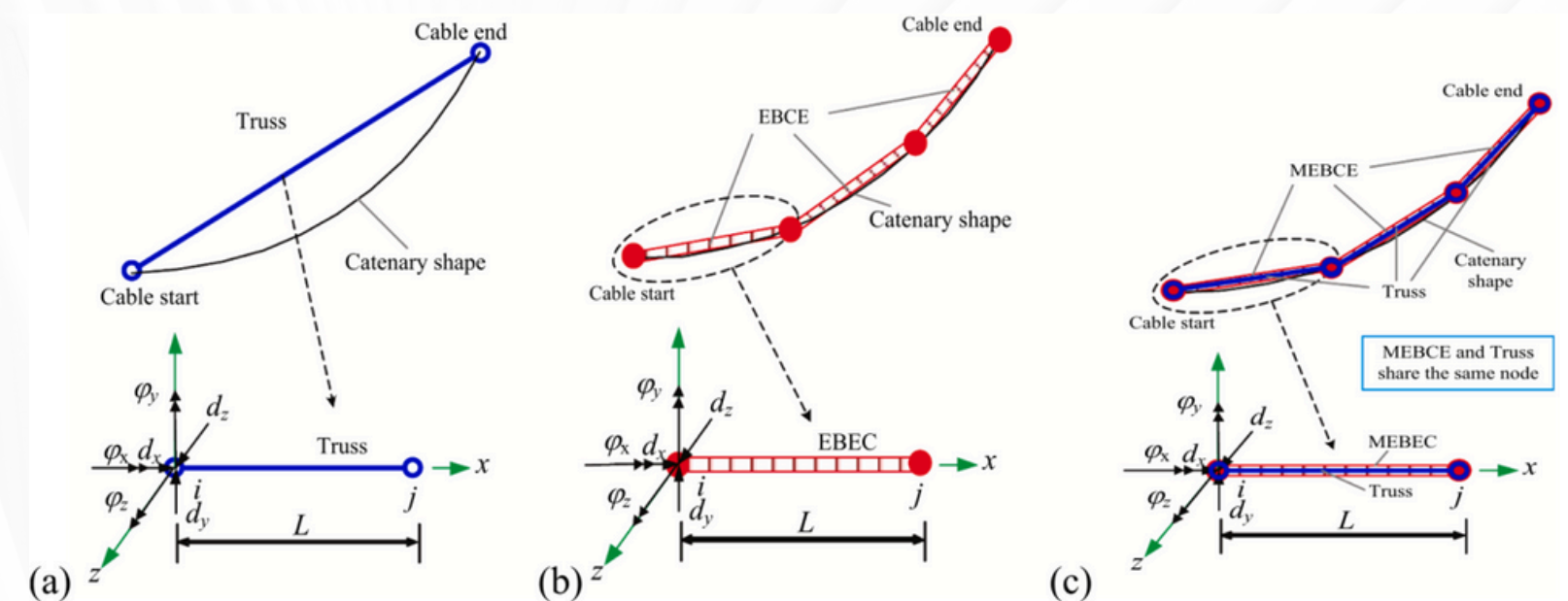
Background

Background significance of the topic

- Damping is essential for fatigue control in long cables.
- Coupled truss-beam modeling gives realistic dynamics.
- Discrete modeling effectively simulates nonlinear damping.
- Our project builds on these works by adopting a discrete-node model to study viscous damping mechanisms in a single inclined stay cable.



Sun (2022), Adv. Struct. Eng., 25(16), 3368–3404



Yi (2023), Eng. Struct., 277, 115443

Objectives

Proposed Approach and Anticipated Contributions

A. Discrete Dynamic Model

B. Simplified Aerodynamic Model

C. Quantitative Assessment of Nonlinearity and Damping

D. Integration with real data

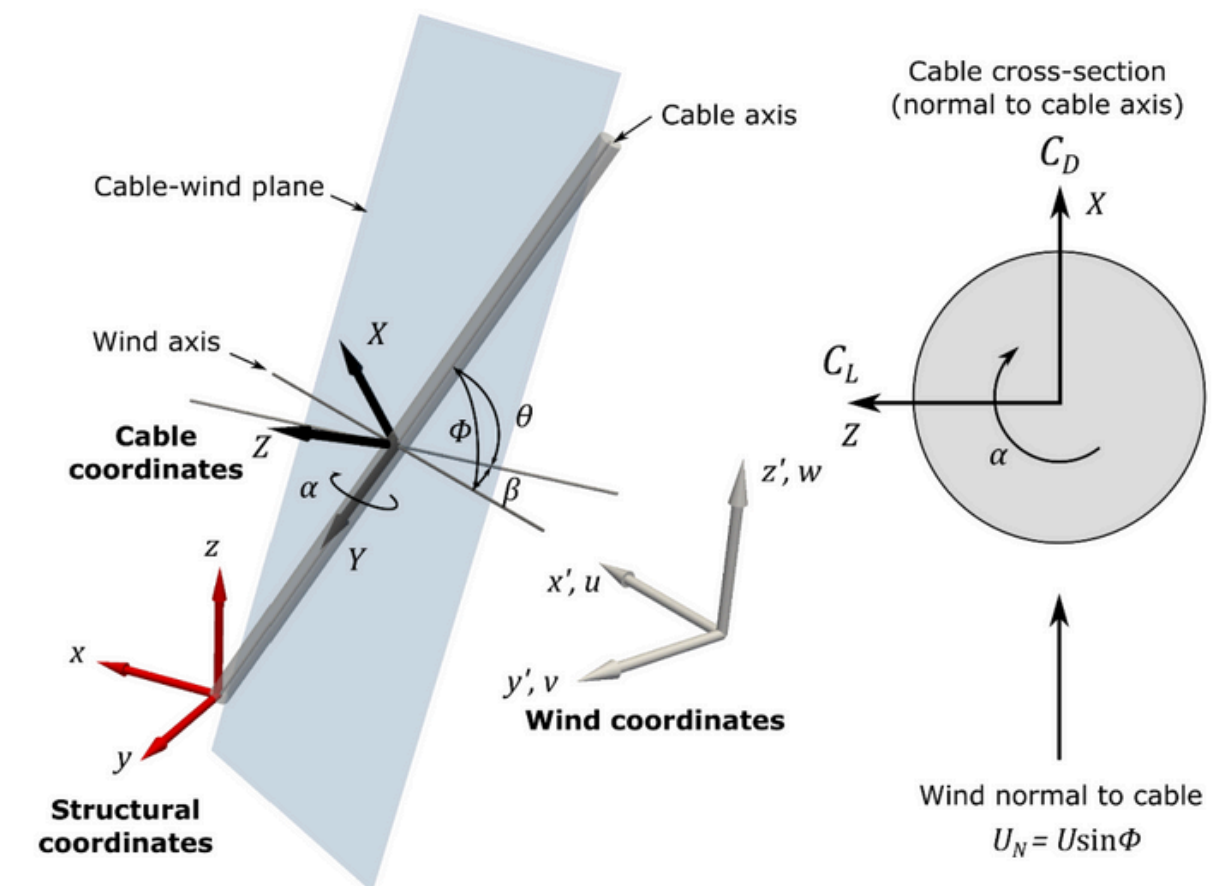
$$\begin{aligned}\zeta_{aero} = \frac{\mu Re}{4m\omega_n} & \left\{ \cos \alpha \left[C_D \left(2 \sin \Phi \cos \alpha + \frac{\sin^2 \alpha}{\cos \alpha \sin \Phi} \right) + \frac{\partial C_D}{\partial Re} Re \sin \Phi \cos \alpha \right. \right. \\ & + \left. \frac{\partial C_D}{\partial \Phi} \cos \Phi \cos \alpha - \frac{\partial C_D}{\partial \alpha_0} \frac{\sin \alpha}{\sin \Phi} \right] \\ & - \sin \alpha \left[C_L \left(2 \sin \Phi \cos \alpha - \frac{\cos \alpha}{\sin \Phi} \right) + \frac{\partial C_L}{\partial Re} Re \sin \Phi \cos \alpha \right. \\ & + \left. \left. \frac{\partial C_L}{\partial \Phi} \cos \Phi \cos \alpha - \frac{\partial C_L}{\partial \alpha_0} \frac{\sin \alpha}{\sin \Phi} \right] \right\}\end{aligned}$$

Where:

μ = air dynamic viscosity.

ζ_{aero} = aerodynamic damping (ratio of critical).

ω_n = circular frequency of the nth cable mode.



Source: FHWA.

- [1] Sun, L. Chen, and H. Huang, “Stay cable vibration mitigation: A review,” *Adv. Struct. Eng.*, vol. 25, no. 16, pp. 3368–3404, Dec. 2022, doi: 10.1177/13694332221132316.
- [2] D. T. Nguyen, S. Q. Chau, and H. N. Phan, “Damping effects of a stay cable attached with a viscous damper and a high damping rubber damper considering cable bending stiffness and damper support flexibility,” *Adv. Bridge Eng.*, vol. 5, no. 1, p. 12, May 2024, doi: 10.1186/s43251-024-00123-7.
- [3] J. Yi, “Modeling and analysis of cable vibrations in cable-stayed bridges under near-fault ground motions,” *Eng. Struct.*, vol. 277, p. 115443, Feb. 2023, doi: 10.1016/j.engstruct.2022.115443.
- [4] P. Egger and L. Caracoglia, “New discrete mass model examining the dynamics of multiple-element-pendulum impact dampers for bridge stay-cables,” *Eng. Struct.*, vol. 253, p. 113665, Feb. 2022, doi: 10.1016/j.engstruct.2021.113665.
- [5] Federal Highway Administration, *Aerodynamic Stability of Bridge Stay Cables—Dynamic Tests and Simulations* (Washington, DC: 2022) <https://highways.dot.gov/sites/fhwa.dot.gov/files/FHWA-HRT-22-097.pdf>